

KAUSTUBH VIHIRKAR

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PROFESSIONAL SUMMARY

AI Engineer with 4+ years of experience building and deploying machine learning and deep learning models. Proficient in Python, TensorFlow, PyTorch, and a range of neural network architectures including CNNs, RNNs, GANs, LLMs, and transformers. Extensive experience optimizing models for production, fine-tuning pretrained networks, and working with large-scale datasets. Skilled in integrating AI solutions into cloud-based platforms and Docker environments, with a strong foundation in data science and algorithm development. Passionate about leveraging AI to solve real-world problems and continuously improving performance through innovation and collaboration.

WORK EXPERIENCE

AI Engineer | GOML

March 2026 – Present

DRL-Based Document Matching

- Built a document-matching system using Deep Reinforcement Learning (DRL), where users upload documents to Amazon S3 and incoming requests are matched against the uploaded document content.
- Designed a reward-based matching pipeline that continuously improves retrieval relevance based on match quality feedback.
- Built a scalable document ingestion and storage pipeline on Amazon S3 to support real-time matching.
- Deployed the matching engine to production, enabling real-time request-to-document matching for end users.

RAG Chatbot for Uploaded Documents

- Developed a Retrieval-Augmented Generation (RAG) chatbot that answers user queries grounded in their uploaded documents.
- Implemented document chunking, embedding generation, and vector-based retrieval to ground chatbot responses in source content.
- Integrated LLM-based response generation with retrieved context to produce accurate, document-grounded answers.
- Enabled a seamless workflow for users to upload documents and immediately query them through the chatbot.

Software Engineer AIML | Brainvire, Ahmedabad

April 2025 – Present

Shopping Assistance Chatbot (Agentic AI)

- Developed a multimodal AI-powered shopping virtual assistant capable of responding in both text and voice.
- Integrated OpenAI GPT-4.0 Mini to handle natural language understanding and generate accurate, context-aware responses.
- Implemented Whisper AI for high-accuracy voice-to-text processing, enabling customers to interact via voice input.
- Added a text-to-speech system to deliver natural-sounding voice responses from the chatbot.
- Connected the assistant to a customer database to retrieve personalized user information.
- Developed and exposed all chatbot functionalities through custom Django-based APIs for scalable integration, supporting product inquiries, order tracking, policy questions, and general customer support.

Book Cover Generation (E-commerce)

- Designed an automated book cover generation pipeline using the Gemini image generation model, turning a short summary and author name into visually aligned cover art.
- Built logic to extract theme, mood, tone, genre, and key visual elements from the summary to inform cover design.
- Implemented intelligent selection of color palettes, art styles, and typography based on book context, generating multiple design variations per book for author review.

AI Engineer | PixelPhant, Udaipur

July 2023 – April 2025

Image Segmentation

- Developed and fine-tuned binary and multiclass image segmentation models using UNet, ISNet, U2Net, Attention UNet, and UNet++ for applications such as e-commerce imaging and background removal.
- Applied advanced attention mechanisms to refine segmentation of complex regions, improving boundary detection accuracy.

- Implemented custom loss functions and data augmentation techniques to enhance model performance, and used morphological post-processing to reduce noise and refine edges.

Agentic AI Financial Chatbot

- Developed an AI-driven financial chatbot using Agentic AI and Groq's API to provide real-time financial insights and assistance.
- Designed advanced NLP techniques for contextual understanding, ensuring accurate and efficient responses.
- Optimized model performance and deployed the chatbot for enterprise use, ensuring seamless integration into the client's financial ecosystem with detailed handover documentation.

Image Generation

- Developed and tested image generation pipelines using pix2pix and LaMa models for background, skin, and apparel retouching.
- Trained pix2pix models on custom datasets to generate high-quality outputs from low-resolution inputs, and used LaMa for mask-based image completion.
- Optimized model performance and reduced inference time through architectural modifications and GPU-based acceleration, deploying models via Docker.

Image Detection

- Developed and optimized image detection models using YOLOv8 to detect product categories such as tops, bottoms, shoes, humans, and e-commerce products.
- Fine-tuned pre-trained YOLO models on in-house datasets and optimized inference speed through pruning and quantization.
- Deployed YOLO models to production using Docker and cloud platforms (E2E/GCP).

Image Labeling

- Implemented automated image labeling pipelines using Grounding DINO for object detection and Segment Anything Model (SAM) for pixel-level segmentation, significantly reducing manual annotation effort.
- Developed custom scripts to automate labeling and segmentation of large-scale image datasets and deployed inference via Docker.

ML Engineer | Value C Consulting Services, Gurgaon

October 2021 – June 2023

Fashion Recommendation

- Developed and deployed a personalized fashion recommendation system using pretrained deep learning models (DenseNet121, ResNet50) to recommend clothing and accessories.
- Leveraged image recognition and classification to extract features from product images, enabling recommendations of visually similar items based on style, color, and trends.
- Collaborated with fashion experts, data scientists, and the marketing team to align recommendation strategies with current trends, and refined algorithms using user behavior data.

Sales Forecasting Using Machine Learning

- Developed and deployed an AI-driven sales forecasting model to predict future sales trends with high accuracy.
- Implemented time-series forecasting techniques, including ARIMA, LSTMs, and Prophet, to analyze historical sales data.
- Optimized model performance through hyperparameter tuning and feature engineering, and built an interactive Plotly/Dash dashboard for real-time visualization of sales predictions.

EDUCATION

- MBA — IT and Business Analytics
- B.E. — Electrical Engineering

SKILLS

Technical Skills: Deep Learning, Machine Learning, GANs, Reinforcement Learning (DRL), Retrieval-Augmented Generation (RAG), Image Segmentation, Image Detection, KNN, Recommendation Systems

Tools & Software: Linux, Docker, AWS S3, E2E Dashboard, GCP Dashboard, Photopea/Photoshop, NAS, FileZilla, VS Code, PyCharm